

TID Taxable Income Cheat

1	125K	No
2	100K	No
3	70K	No
4	120K	No
5	95K	Yes
6	60K	No
7	220K	No
8	85K	Yes
9	75K	No
10	90K	Yes

Sorted values:

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6	60K	No
3	70K	No
9	75K	No
8	85K	Yes
10	90K	Yes
5	95K	Yes
2	100K	No
4	120K	No
1	125K	No
7	220K	No

Step 2: Find midpoints between consecutive unique values

Example:

- Between 60K and 70K → midpoint = 65K
- Between 70K and 75K → midpoint = 72.5K
- ...and so on.

This gives the list of possible split points **M**.

Step 3: For each midpoint v , count N_{vi}

For each v :

- N_{vi} for "No" = number of "No" with Taxable Income $\leq v$
- N_{vi} for "Yes" = number of "Yes" with Taxable Income $\leq v$

Example:

For $v = 72.5K$:

- "No" with $\leq 72.5K$: {60K, 70K} → 2
- "Yes" with $\leq 72.5K$: {} → 0

Step 4: Calculate probabilities

Using:

$$\hat{P}(c_i|DY) = \frac{N_{vi}}{\sum_{j=1}^k N_{vj}}$$
$$\hat{P}(c_i|DN) = \frac{n_i - N_{vi}}{\sum_{j=1}^k (n_j - N_{vj})}$$

Loop j = 1 to n-1 → counts for building midpoints and N_{vi}

Extra y_n check → fixes n_i so it's the *total class count*, which is required for computing the **right-side group DN** stats.

Step 5: Calculate the Gain

For each v :

$$\text{Gain}(D, DY, DN) = H(D) - \frac{|DY|}{|D|} H(DY) - \frac{|DN|}{|D|} H(DN)$$

- H = entropy formula.

Step 6: Pick the best split point

The algorithm compares all split points' gains and picks the one with **maximum gain** (or **minimum Gini index**, if using Gini).